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Graded assignment • 2h

Your grade: 90%

Your latest: 90% • Your highest: 90% • To pass you need at least 80%. We keep your highest score.

1. Which of the following are valid parameters for the `Box.from_directory` function in Keras' `ImageDataGenerator`? 1 / 1 point

- rotation_range
- target_size
- Batchnormalize_center
- batch_size
- class_mode
- validation_split

Correct
`target_size` specifies the dimensions to which all images found will be resized.

Correct
`batch_size` determines the number of images to be yielded from the generator per batch.

Correct
`class_mode` determines the type of label arrays that are returned: `'categorical'`, `'binary'`, `'sparse'`, etc.

2. What are the advantages of using Keras Tuner for hyperparameter optimization? 1 / 1 point

- Automates the tuning process
- Eliminates the need for domain expertise
- Always provides the best model
- Provides a systematic approach for hyperparameter search
- Supports multiple search algorithms
- Keras Tuner supports various search algorithms to optimize hyperparameters.

Correct
Automating the tuning process can save time and effort.

Correct
Best done! Keras Tuner provides a structured way to conduct hyperparameter search.

Correct
Best done! Constraint image size is essential for accurate predictions as VGG 16 expects images of a specific dimension.

3. Which of the following is a key component when defining a convolutional Neural Network (CNN) model? 1 / 1 point

- Number of feature maps
- Batch size
- Epochs
- Learning rate

Correct
The number of feature maps is a crucial component in a CNN as it determines the depth of the output of a convolutional layer.

4. Which of the following statements correctly describe transfer learning and its application in deep learning? 1 / 1 point

- Transfer learning is useful especially when dealing with limited labeled data.
- Transfer learning requires training a new model from scratch for every new task.
- Transfer learning involves using a pre-trained model on a new, related task.

Correct
Correct! Transfer learning leverages pre-trained models to improve performance on related tasks.

Correct
Correct! By using pre-trained models, transfer learning can save significant time and computational resources.

- Using transfer learning, models can be fine-tuned for specific tasks with relatively small datasets.

Correct
Correct! Fine-tuning a pre-trained model with a small dataset is a common practice in transfer learning.

5. Which of the following statements accurately describe data augmentation and visualization? 1 / 1 point

- A digital image is represented as a set of numbers corresponding to pixel values.

Correct
Correct! Digital images are indeed represented as a set of numbers, where each number corresponds to a pixel value.

- The pixel values in grayscale images are always in binary format.
- All pixels in a digital image must have the same value.
- Pixels in color images can have multiple values to represent different colors.

Correct
Correct! Each pixel in color images can have multiple values, typically representing the intensity of red, green, and blue components.

- Pixels are the basic unit of any digital image.

Correct
Correct! Pixels are indeed the basic unit of any digital image.

6. Which of the following plots is best suited for visualizing the distribution and spread of continuous data? 1 / 1 point

- Heat map
- KDE plot
- Count plot
- Box plot

Incorrect
Almost. A KDE plot shows the probability density of continuous data, but it doesn't explicitly show the spread and quartiles like a box plot does.